STAT 626 Homework 2

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1.

2.2)

We have $x_t = \beta_0 + \beta_1 t + w_t$ where w_t is white noise, so mean of 0 and variance of σ_w^2.

a)

In order for x_t to be stationary we must have $E[x_t]$ constant that does not depend on t. And $Cov(x_t, x_s)$ only depends on the lag $|t - s|$ and not the actual values of t and s.

We start with the first condition.

$$E[x_t] = \beta_0 + \beta_1 t$$

This clearly depends on t, so x_t is not stationary.

b)

We have $y_t = x_t - x_{t-1}$ when we rewrite this we have:

$$y_t = x_t - x_{t-1} = \beta_0 + \beta_1 t + w_t - (\beta_0 + \beta_1(t-1) + w_{t-1}) = \beta_1 + w_t - w_{t-1}$$

We start with the first condition of stationary:

$$E[y_t] = \beta_1$$

Since this is constant and does not depend on t, this condition is met.

Next we need to show that $Cov(y_t, y_s)$ only depends on the lag $|t - s|$:

$$Cov(y_t, y_s) = Cov(\beta_1 + w_t - w_{t-1}, \beta_1 + w_s - w_{s-1})$$

When $s = t$ we have:

$$Cov(y_t, y_t) = Var(y_t) = Var(\beta_1 + w_t - w_{t-1}) = 2\sigma_w^2$$

When $s = |t-1|$ we have:

$$Cov(y_t, y_{t-1}) = Cov(\beta_1 + w_t - w_{t-1}, \beta_1 + w_{t-1} - w_{t-2}) = Cov(-w_{t-1}, w_{t-1}) = -Var(w_{t-1}) = -\sigma_w^2$$

When $s = |t-2|$ we have:

$$Cov(y_t, y_{t-2}) = Cov(\beta_1 + w_t - w_{t-1}, \beta_1 + w_{t-2} - w_{t-3}) = 0$$

So we have:

$$\gamma(h) = \begin{cases} 2\sigma_w^2 & h = 0 \\ -\sigma_w^2 & h = 1 \\ 0 & h \geq 2 \end{cases}$$

Since we have shown that the auto covariance function depends on s and t only through the lag h we can say y_t is stationary.

c)

We the two sided moving average $v_t = \frac{1}{3}(x_{t-1} + x_t + x_{t+1})$. To start, we expand out each x:

$$v_t = \frac{1}{3}(\beta_0 + \beta_1(t-1) + w_{t-1} + \beta_0 + \beta_1(t) + w_t + \beta_0 + \beta_1(t+1) + w_{t+1}) = \frac{1}{3}(3\beta_0 + 3\beta_1 + w_{t-1} + w_t + w_{t+1})$$

Then to get the mean we take the expected value:

$$E[v_t] = E[\beta_0 + \beta_1 + w_{t-1} + w_t + w_{t+1}] = \beta_0 + \beta_1$$

2.3)

We have the two sided moving average smoother $x_t = \frac{1}{4}(w_{t-1} + 2w_t + w_{t+1})$ with w_t white noise:

To find the the autocovariance function we start with the case of $h = 0$:

$$\gamma(h) = Cov(x_t, x_t) = Var(\frac{1}{4}(w_{t-1} + 2w_t + w_{t+1})) = \frac{1}{16}(\sigma_w^2 + 4\sigma_w^2 + \sigma_w^2) = \frac{6}{16}\sigma_w^2 = \frac{3}{8}\sigma_w^2$$

Then the case of $h = 1$:

$$\gamma(1) = Cov(x_{t+1}, x_t) = Cov(\frac{1}{4}(w_t + 2w_{t+1} + w_{t+2}), \frac{1}{4}(w_{t-1} + 2w_t + w_{t+1})) = \frac{1}{16}(Cov(w_t, 2w_t) + Cov(2w_{t+1}, w_{t+1}))$$

$$= \frac{1}{16}(2\sigma_w^2 + 2\sigma_w^2) = \frac{1}{4}\sigma_w^2$$

Then the case of $h = 2$:

$$\gamma(2) = Cov(x_{t+2}, x_t) = Cov(\frac{1}{4}(w_{t+1} + 2w_{t+2} + w_{t+3}), \frac{1}{4}(w_{t-1} + 2w_t + w_{t+1})) = \frac{1}{16}Cov(w_{t+1}, w_{t+1}) = \frac{1}{16}\sigma_w^2$$

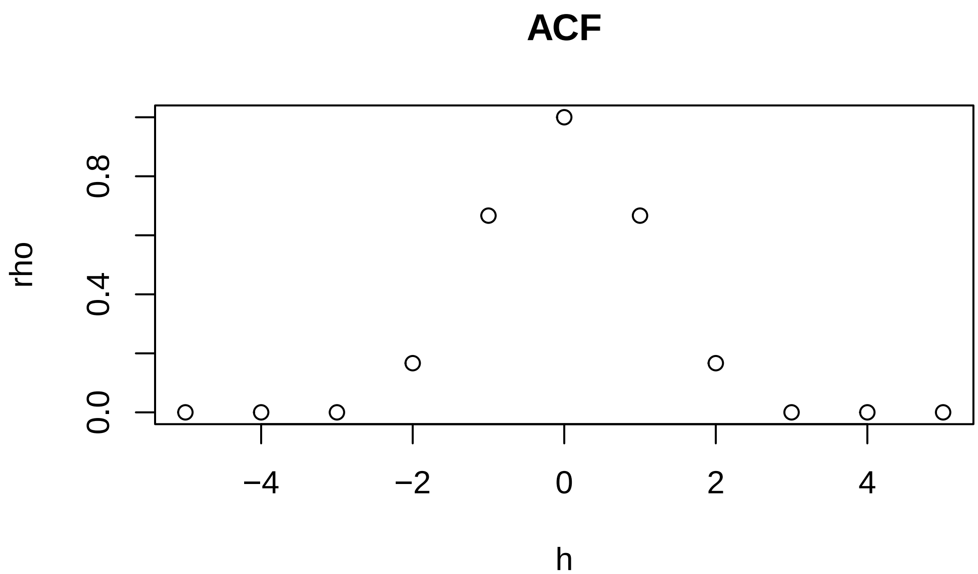
So we have the autocovariance function:

$$\gamma(h) = \begin{cases} \frac{3}{8}\sigma_w^2 & h = 0 \\ \frac{1}{4}\sigma_w^2 & h = 1 \\ \frac{1}{16}\sigma_w^2 & h = 2 \\ 0 & h \geq 3 \end{cases}$$

Then we get the autocorrelation function by dividing each term by $\gamma(0)$ so:

$$\rho(h) = \begin{cases} 1 & h = 0 \\ \frac{2}{3} & h = 1 \\ \frac{1}{6} & h = 2 \\ 0 & h \geq 3 \end{cases}$$

Here is a sketch of the ACF:



2.4)

We have $x_t = \phi x_{t-1} + w_t$ where $w_t \sim wn(0, 1)$ and x_t is stationary and x_{t-1} is uncorrelated with noise w_t.

a)

To show this we use the fact that x_t is stationary, and thus expected value is constant and does not depend on t. So:

$$E[x_t] = E[\phi x_{t-1} + w_t] = \phi E[x_{t-1}]$$

Since x_t is stationary $E[x_t] = E[x_{t-1}]$, thus the only $E[x_t]$ that always has $E[x_t] = \phi E[x_{t-1}]$ is $E[x_t] = \mu_{xt} = 0$.

b)

$\gamma(0) = Var(x_t) = Var(\phi x_{t-1} + w_t) = \phi^2 Var(x_{t-1}) + 1$

We use the fact that x_t is stationary, and thus autocovariance only depends on the lag component and not t so say that in general $Var(x_i) = \gamma(0)$. Therefore $Var(x_{t-1}) = \gamma(0)$:

$$\gamma(0) = \phi^2 \gamma(0) + 1$$

$$\gamma(0) - \phi^2\gamma(0) = 1$$

$$\gamma(0) = \frac{1}{1-\phi^2}$$

c)

Since variance cannot be negative only $\phi < 1$ makes sense.

d)

First we want to find $\gamma(1)$:

$$\gamma(1) = Cov(x_{t+1}, x_t) = Cov(\phi x_t + w_{t+1}, x_t) = Cov(\phi x_t, x_t) = \phi Var(x_t) = \phi\gamma(0)$$

Then we can find the log-one autocorrelation easily:

$$\rho(1) = \frac{\gamma(1)}{\gamma(0)} = \frac{\phi\gamma(0)}{\gamma(0)} = \phi$$

2.5)

We have $x_t = \delta + x_{t-1} + w_t$ with $x_0 = 0$ and $w_t \sim wn(0, \sigma_w^2)$.

a)

To show this we start expanding x_{t-1}:

$$x_t = \delta + \delta + x_{t-2} + w_{t-1} + w_t = \delta + \delta + \delta + x_{t-3} + w_{t-2} + w_{t-1} + w_t = \delta t + \sum_{k=1}^{t} w_k$$

b)

$$E[x_t] = E[\delta t] + \sum_{k=1}^{t} w_k] = \delta t$$

To find autocovariance I start with $\gamma(t, t)$:

$$\gamma(0) = Var(x_t) = Var(\delta t + \sum_{k=1}^{t} w_k) = t\sigma_w^2$$

Then $\gamma(t+1,t)$:

$$\gamma(t+1,t) = Cov(x_{t+1}, x_t) = Cov(\delta(t+1) + \sum_{k=1}^{t+1} w_k, \delta t + \sum_{k=1}^{t} w_k) = t\sigma_w^2$$

The pattern becomes clear, the number of matching terms is what determines the coefficient in front of σ_w^2. Or in other terms:

$$\gamma(s,t) = \min\{s,t\}\sigma_w^2$$

c)

Stationary series have constant expected value that does not depend on t. $E[x_t] = \delta t$ explicitly depends on t.

d)

$$p_x(t-1,t) = \sqrt{\frac{t-1}{t}}, \lim_{t\to\inf}(\frac{t-1}{t}) = 1, \text{ so } \lim_{t\to\inf}\sqrt{\frac{t-1}{t}} = 1$$

This suggests that as we look farther and farther into the future, points that are near each other are related more and more.

e)

I would use the transformation $y_t = x_t - \delta - x_{t-1}$. To prove this is stationary we start with its expected value:

$$E[y_t] = E[x_t - \delta - x_{t-1}] = E[\delta + x_{t-1} + w_t - \delta - x_{t-1}] = E[w_t] = 0$$

Expected value is constant and does not depend on t.

Next we look at autocovariance, starting with $\gamma(0)$:

$$\gamma(0) = Var(y_t) = Var(w_t) = \sigma_w^2$$

Next $\gamma(1)$:

$$\gamma(1) = Cov(y_{t+1}, y_t) = Cov(x_{t+1} - \delta - x_t, x_t - \delta - x_{t-1}) = Cov(w_{t+1}, w_t) = 0$$

So we have the autocovariance function of:

$$\gamma(h) = \begin{cases} \sigma_w^2 & h = 0 \\ 0 & h \geq 1 \end{cases}$$

Since autocovariance only depends on lag, this is a stationary series.

2.6)

I would treat this data as non stationary. The first necessary aspect of a stationary series is that expected value is constant and does not depend on t. There is an increasing trend in the data, as t increases the values of temperature deviations are increasing. So the expected value of x_t depends on t.

2.9)

We have $x_t = w_t w_{t-1}$ with w_t as white noise.

$$E[x_t] = E[w_t w_{t-1}]$$

Since w_t is a white noise process, each w_i is independent:

$$E[w_t w_{t-1}] = E[w_t]E[w_{t-1}] = 0$$

Starting with $\gamma(t,t)$:

$$\gamma(t,t) = Var(x_t) = Var(w_t w_{t-1}) = E[(w_t w_{t-1} - \mu_{x_t})^2] = E[(w_t w_{t-1})^2] = E[w_t^2]E[w_{t-1}^2] = \sigma_w^4$$

Then $\gamma(t+1,t)$:

$$\gamma(t+1,t) = Cov(x_{t+1}, x_t) = Cov(w_{t+1}w_t, w_t w_{t-1})$$

$$= E[(w_{t+1}w_t)(w_t w_{t-1})] = E[w_{t+1}w_t^2 w_{t-1}] = E[w_{t+1}]E[w_t^2]E[w_{t-1}] = 0$$

And it will continue being 0 as lag increases so:

$$\gamma(h) = \begin{cases} \sigma_w^4 & h = 0 \\ 0 & h \geq 1 \end{cases}$$

Since expected value is constant and does not depend on t and the autocovariance function only depends on lag, we can say that x_t is stationary.

2.

We have X_1 with $Var(X_1) = 1$ and X_2 with $Var(X_2) = 4$, and $Cov(X_1, X_2) = -2$.

a)

$Var(X_1 - 5X_2) = Var(X_1) + 25Var(X_2) - 2(5Cov(X_1, X_2)) = 1 + (25)(4) - (2)(5)(-2) = 121$

$Var(X_1 + 5X_2) = Var(X_1) + 25Var(X_2) + 2(5)(Cov(X_1, X_2)) = 1 + (25)(4) + (2)(5)(-2) = 81$

$Var(X_1 + X_2) = Var(X_1) + Var(X_2) + 2Cov(X_1, X_2) = 1 + 4 + (2)(-2) = 1$

b)

$Var(X_1 - cX_2) = Var(X_1) + c^2 Var(X_2) - 2cCov(X_1, X_2)$

Take partial derivative relative to c:

$$\frac{\delta}{\delta c} = 2cVar(X_2) - 2Cov(X_1, X_2)$$

Set equal to zero:

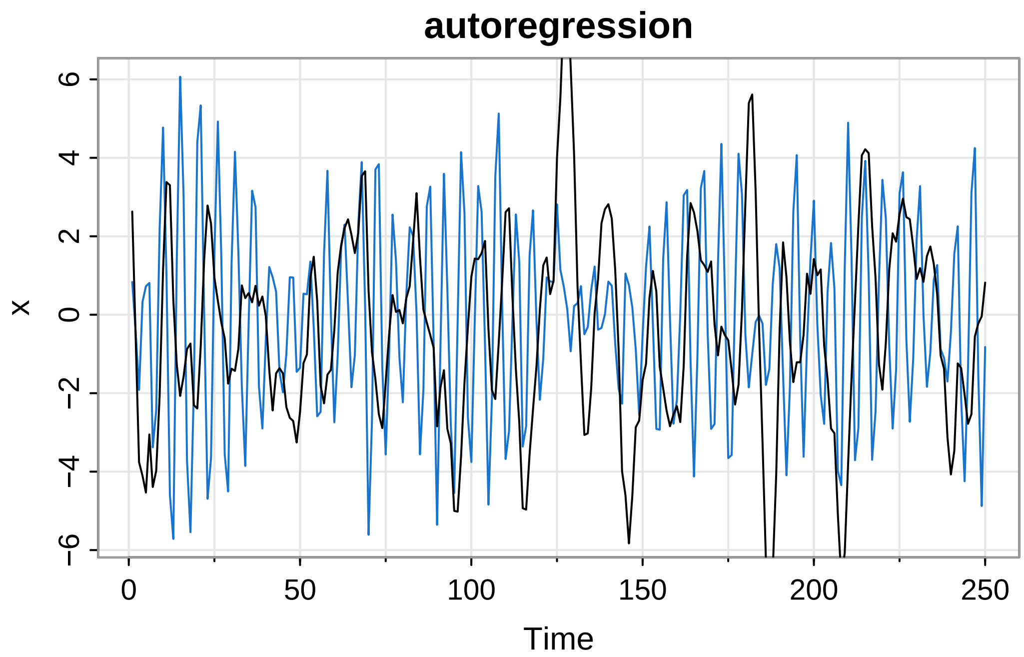
$$2cVar(X_2) = 2Cov(X_1, X_2)$$

$$c = Cov(X_1, X_2)/Var(X_2) = \frac{-2}{4} = -1/2$$

3.

My UIN starts with 6 and ends with 9 so we have the model:

$$x_t = .6x_{t-1} - .9x_{t-2} + w_t$$



Blue is the UIN series, black is the original series.

NO

b)

These two time series have similar magnitude and do not seem to have drift as both vary around 0.

However they differ, the time series with my UIN digits as coefficients has faster oscillations while the other time series is smoother and looks like it has more defined cycles.

4.

We have $x_t = w_{t-1}w_{t-2}(w_t + t)$ with $w_t \sim i.i.d.N(0, \sigma_w^2)$.

$$E[x_t] = E[w_{t-1}w_{t-2}(w_t+t)] = E[w_{t-1}w_{t-2}w_t]+E[w_{t-1}w_{t-2}t] = E[w_{t-1}]E[w_{t-2}]E[w_t]+tE[w_{t-1}]E[w_{t-2}] = 0$$

To find the autocovariance function we start with $\gamma(t, t)$:

$$\gamma(t+1, t) = Var(x_t) = Var(w_{t-1}w_{t-2}(w_t + t)) = Var(w_{t-1}w_{t-2}w_t + w_{t-1}w_{t-2}t)$$

$$Var(w_{t-1}w_{t-2}w_t) + Var(w_{t-1}w_{t-2}t) + 2Cov(w_{t-1}w_{t-2}w_t, w_{t-1}w_{t-2}t)$$

$$E[w_{t-1}^2 w_{t-2}^2 w_t^2] + t^2 E[w_{t-1}^2 w_{t-2}^2] + 2tE[w_{t-1}^2 w_{t-2}^2 w_t]$$

$$= \sigma_w^6 + t^2 \sigma_w^4$$

Next $\gamma(t+1, t)$:

$$\gamma(t+1, t) = Cov(w_t w_{t-1}(w_{t+1}(t+1)), w_{t-1}w_{t-2}(w_t + t))$$

$$= Cov(w_t w_{t-1} w_{t+1} + w_t w_{t-1}(t+1), w_{t-1}w_{t-2}w_t + w_{t-1}w_{t-2}t)$$

$$= Cov(w_t w_{t-1} w_{t+1}, w_{t-1}w_{t-2}w_t) + Cov(w_t w_{t-1} w_{t+1}, w_{t-1}w_{t-2}t)$$

$$+Cov(w_t w_{t-1}(t+1), w_{t-1}w_{t-2}w_t) + Cov(w_t w_{t-1}(t+1), w_{t-1}w_{t-2}t)$$

$$= E[w_t^2 w_{t-1}^2 w_{t+1} w_{t-2}] + tE[w_t w_{t-1}^2 w_{t+1} w_{t-2}] + (t+1)E[w_t^2 w_{t-1}^2 w_{t-2}] + t(t+1)E[w_t w_{t-1}^2 w_{t-2}]$$

Each of these terms have at least one $E[w_i]$, since $E[w_i]$ we end up with zero. So the autocovariance function is:

$$\gamma(t, s) = \begin{cases} \sigma_w^6 + t^2\sigma_w^4 & s = t \\ 0 & s \neq t \end{cases}$$

We can say that x_t is not stationary because the covariance function depends on the actual value t, and not the lag $|s - t|$.